



Technical Service Bulletin

Technical Service Bulletin: TSB160116

Released Date: 04-Nov-2016

New Piston Ring Compressor

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- ISX (All 15 Liter Products)
- QSX (All 15 Liter Products)
- X15 (All Versions)

Issue

When installing piston and connecting rod assemblies, the piston is **not** held securely in piston ring compressor. Reduced tension piston ring packs offer less resistance to piston movement within piston ring compressor.

Resolution

- A new piston ring compressor part number has been released that incorporates a pair of plastic tipped thumbscrews to improve piston retention. See Table 1 below.
- This document also authorizes repair locations to modify existing piston ring compressors by drilling and tapping two holes for installation of plastic tipped thumbscrews. See Service Instructions section below.
 - Modification to existing piston ring compressors can be made in-house or by a machine shop.
 - Labor to modify piston ring compressor tool, Part Number 3164846, will **not** be reimbursed by Cummins Inc.

Service Tool Availability

Service tools are available. See Table 1 for part numbers.

Table 1, Service Tools				
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Piston Ring Compressor	3164846	Yes	Yes	5394454
Thumbscrew Assembly	-	-	-	5394447*
*Quantity of two required only if modifying existing part number 3164846.				

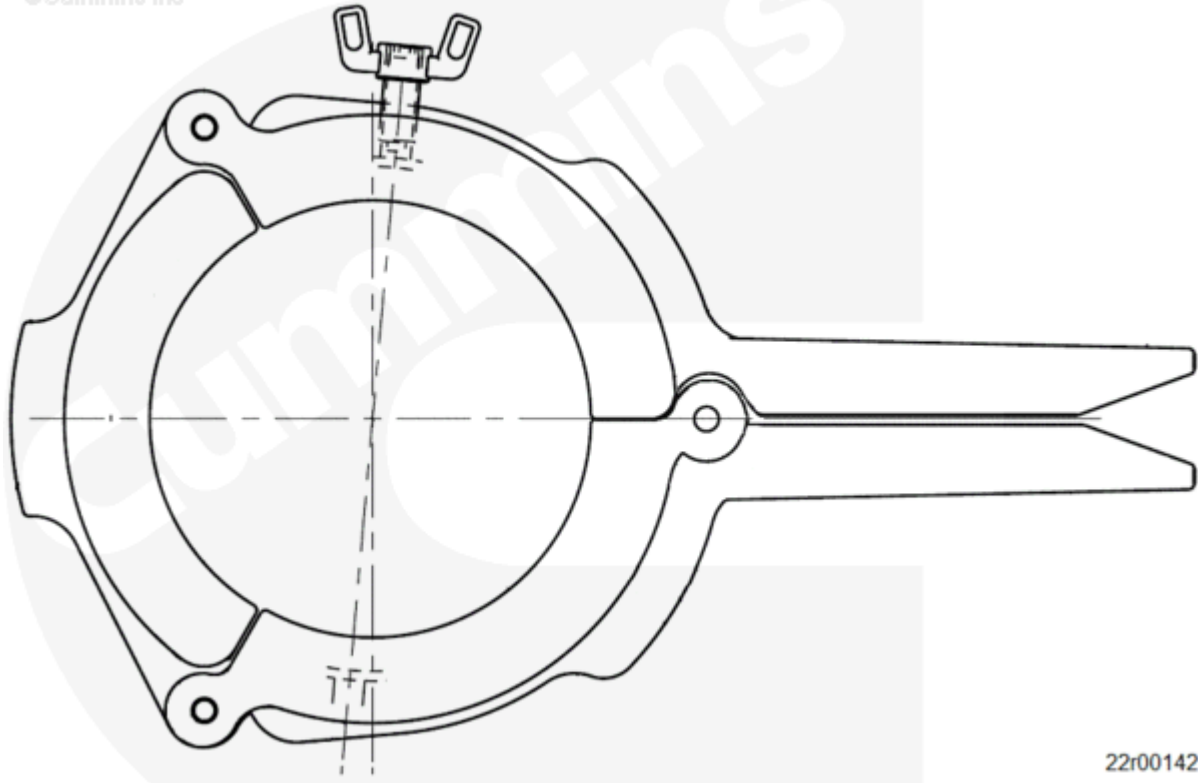
Service Instructions

Piston Ring Compressor, Part Number 3164846, Modification Instructions:

⚠ WARNING ⚠

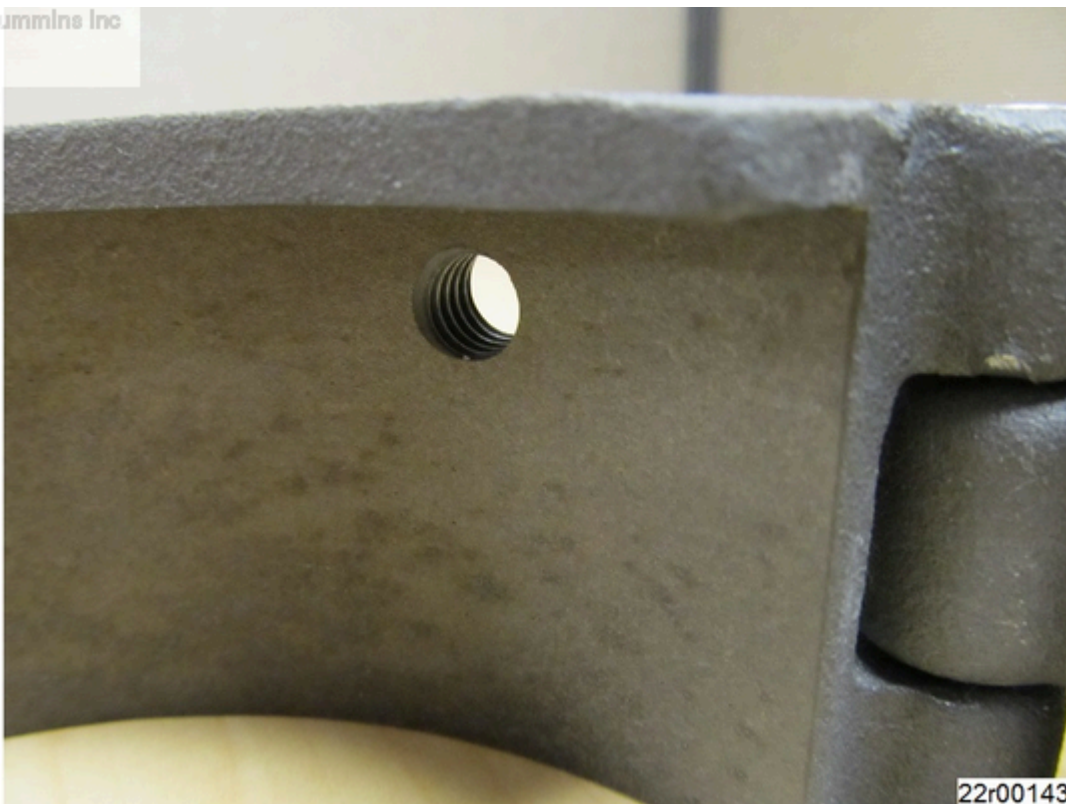
To reduce the possibility of personal injury, wear goggles and protective clothing.

1. Locate area for holes:
 - Holes **must** be approximately 90 degrees from handle centerline and 180 degrees apart. See Figure 1 below.
 - Holes **must** be 8.84 mm [11/32 in] down from inside of upper ledge of piston ring compressor for compatibility with high ring pistons. See Figure 2 below.
2. Center punch locations for holes on inside of piston ring compressor bore.
3. Drill holes with a 7.94 mm [5/16 in] drill bit.
4. Tap holes with a 3/8-16 unified thread standard (UNC) tap.
5. Verify no burrs extend into bore.
6. Install two thumbscrews, Part Number 5394447. See Figures 3 and 4 below.



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Figure 1, Location of Thumbscrew Holes.



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Figure 2, Location of Thumbscrew Holes on Piston Ring Compressor to be Drilled and Tapped.



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Figure 3, Thumbscrew, Part Number 5394447.



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Figure 4, Assembled Piston Ring Compressor.

Date	Details
2016-10-27	Module Created
2016-11-3	Added X15 engines.

Last Modified: 04-Nov-2016
